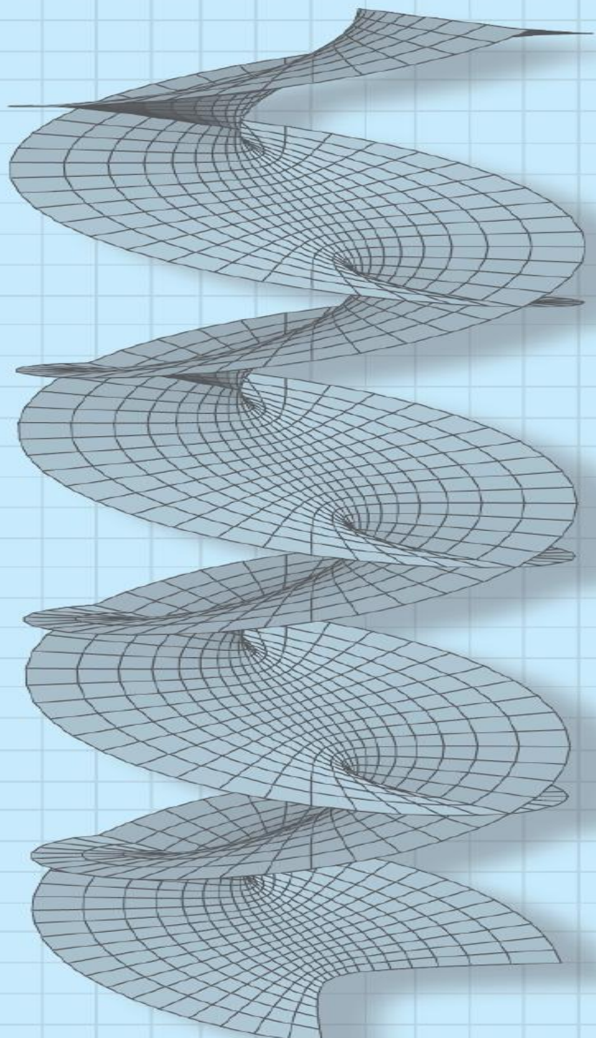

A First Course in

# Complex Analysis

with Applications

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If a complex function $f$ fails to be analytic at a point $z = z_0$, then this point is said to be a singularity or singular point of the function. For example, the complex numbers $z = 2i$ and $z = -2i$ are singularities of the function $f(z) = z/(z^2+4)$ because $f$ is discontinuous at each of these points. [illegible],

In this section we will be concerned with a new kind of "power series" expansion of $f$ about an isolated singularity $z_0$. This new series will involve negative as well as nonnegative integer powers of $z - z_0$.

# Laurent Series

**Isolated Singularities** Suppose that $z = z_0$ is a singularity of a complex function $f$. The point $z = z_0$ is said to be an **isolated singularity** of the function $f$ if there exists *some* deleted neighborhood, or punctured open disk, $0 < |z - z_0| < R$ of $z_0$ throughout which $f$ is analytic. For example, we have just seen that $z = 2i$ and $z = -2i$ are singularities of $f(z) = z/(z^2 + 4)$. Both $2i$ and $-2i$ are isolated singularities since $f$ is analytic at every point

in the neighborhood defined by $|z - 2i| < 1$, except at $z = 2i$, and at every point in the neighborhood defined by $|z - (-2i)| < 1$, except at $z = -2i$. In other words, $f$ is analytic in the deleted neighborhoods $0 < |z - 2i| < 1$ and $0 < |z + 2i| < 1$.

**A New Kind of Series** If $z = z_0$ is a singularity of a function $f$, then certainly $f$ cannot be expanded in a power series with $z_0$ as its center. However, about an isolated singularity $z = z_0$, it is possible to represent $f$ by a series involving both negative and nonnegative integer powers of $z - z_0$; that is,

$$f(z) = \cdots + \frac{a_{-2}}{(z-z_0)^2} + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots. \tag{1}$$

As a *very* simple example of (1) let us consider the function $f(z) = 1/(z-1)$. As can be seen, the point $z = 1$ is an isolated singularity of $f$ and consequently the function cannot be expanded in a Taylor series centered at that point. Nevertheless, $f$ can expanded in a series of the form given in (1) that is valid for all $z$ *near* 1:

$$f(z) = \cdots + \frac{0}{(z-1)^2} + \frac{1}{z-1} + 0 + 0\cdot(z-1) + 0\cdot(z-1)^2 + \cdots. \tag{2}$$

The series representation in (2) is valid for $0 < |z-1| < \infty$.

Using summation notation, we can write (1) as the sum of two series

$$f(z) = \sum_{k=1}^{\infty} a_{-k}(z-z_0)^{-k} + \sum_{k=0}^{\infty} a_k(z-z_0)^k. \tag{3}$$

The two series on the right-hand side in (3) are given special names. The part with negative powers of $z - z_0$, that is,

$$\sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} = \sum_{k=1}^{\infty} \frac{a_{-k}}{(z - z_0)^k} \tag{4}$$

is called the **principal part** of the series (1) and will converge for $|1/(z - z_0)| < r^*$ or equivalently for $|z - z_0| > 1/r^* = r$. The part consisting of the nonnegative powers of $z - z_0$,

$$\sum_{k=0}^{\infty} a_k(z - z_0)^k, \tag{5}$$

is called the **analytic part** of the series (1) and will converge for $|z - z_0| < R$. Hence, the sum of (4) and (5) converges when $z$ satisfies both $|z - z_0| > r$

*and* $|z - z_0| < R$ , that is, when $z$ is a point in an annular domain defined by $r < |z - z_0| < R$.

By summing over negative and nonnegative integers, (1) can be written compactly as

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k.$$

### EXAMPLE 1 Series of the Form Given in (1)

The function $f(z) = \dfrac{\sin z}{z^4}$ is not analytic at the isolated singularity $z = 0$ and hence cannot be expanded in a Maclaurin series. However, $\sin z$ is an entire function, and from (13) of Section 6.2 we know that its Maclaurin series,

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \frac{z^9}{9!} - \cdots,$$

converges for $|z| < \infty$. By dividing this power series by $z^4$ we obtain a series for $f$ with negative and positive integer powers of $z$:

$$f(z) = \frac{\sin z}{z^4} = \overbrace{\frac{1}{z^3} - \frac{1}{3!\,z}}^{\text{principal part}} \overbrace{+ \frac{z}{5!} - \frac{z^3}{7!} + \frac{z^5}{9!} - \cdots}^{\text{analytic part}} . \tag{6}$$

The analytic part of the series in (6) converges for $|z| < \infty$. (Verify.) The principal part is valid for $|z| > 0$. Thus (6) converges for all $z$ except at $z = 0$; that is, the series representation is valid for $0 < |z| < \infty$.

A series representation of a function $f$ that has the form given in (1), and (2) and (6) are such examples, is called a **Laurent series** or a **Laurent expansion** of $f$ about $z_0$ on the annulus $r < |z - z_0| < R$.

**Theorem** **Laurent's Theorem**

Let $f$ be analytic within the annular domain $D$ defined by $r < |z - z_0| < R$. Then $f$ has the series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k \tag{7}$$

valid for $r < |z - z_0| < R$. The coefficients $a_k$ are given by

$$a_k = \frac{1}{2\pi i} \oint_C \frac{f(s)}{(s - z_0)^{k+1}}\, ds, \quad k = 0,\ \pm 1,\ \pm 2,\ \ldots, \tag{8}$$

where $C$ is a simple closed curve that lies entirely within $D$ and has $z_0$ in its interior.

## EXAMPLE 2 Four Laurent Expansions

Expand $f(z) = \dfrac{1}{z(z-1)}$ in a Laurent series valid for the following annular domains.

**(a)** $0 < |z| < 1$ **(b)** $1 < |z|$ **(c)** $0 < |z-1| < 1$ **(d)** $1 < |z-1|$

**Solution** The four specified annular domains are shown in Figure 6.8. The black dots in each figure represent the two isolated singularities, $z = 0$ and $z = 1$, of $f$. In parts (a) and (b) we want to represent $f$ in a series involving only negative and nonnegative integer powers of $z$, whereas in parts (c) and (d) we want to represent $f$ in a series involving negative and nonnegative integer powers of $z - 1$.

(a) By writing

$$f(z) = -\frac{1}{z}\,\frac{1}{1-z},$$

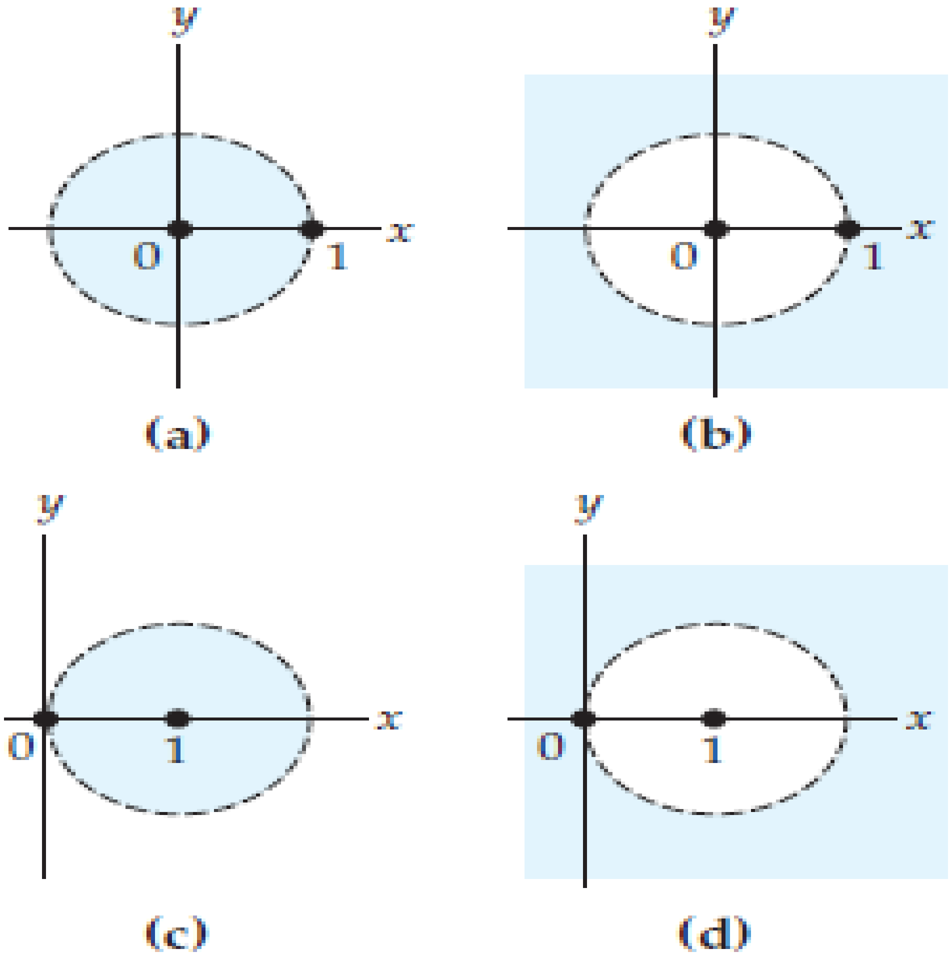
to write $1/(1-z)$ as a series:

$$f(z) = -\frac{1}{z}\left[1 + z + z^2 + z^3 + \cdots\right].$$

The infinite series in the brackets converges for $|z| < 1$, but after we multiply this expression by $1/z$, the resulting series

$$f(z) = -\frac{1}{z} - 1 - z - z^2 - z^3 - \cdots$$

converges for $0 < |z| < 1$.




Annular domains for Example 2

(b) To obtain a series that converges for $1 < |z|$, we start by constructing a series that converges for $|1/z| < 1$. To this end we write the given function $f$ as

$$f(z) = \frac{1}{z^2} \frac{1}{1 - \frac{1}{z}}$$

and with $z$ replaced by $1/z$:

$$f(z) = \frac{1}{z^2}\left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \ldots\right].$$

The series in the brackets converges for $|1/z| < 1$ or equivalently for $1 < |z|$. Thus the required Laurent series is

$$f(z) = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} + \ldots.$$

(c) This is basically the same problem as in part (a), except that we want all powers of $z-1$. To that end, we add and subtract 1 in the denominator with $z$ replaced by $z-1$:

$$\begin{aligned} f(z) &= \frac{1}{(1-1+z)(z-1)} \\ &= \frac{1}{z-1}\,\frac{1}{1+(z-1)} \\ &= \frac{1}{z-1}\left[1-(z-1)+(z-1)^2-(z-1)^3+\cdots\right] \\ &= \frac{1}{z-1}-1+(z-1)-(z-1)^2+\cdots. \end{aligned}$$

The requirement that $z \neq 1$ is equivalent to $0<|z-1|$, and the geometric series in brackets converges for $|z-1|<1$. Thus the last series converges for $z$ satisfying $0<|z-1|$ *and* $|z-1|<1$, that is, for $0<|z-1|<1$.

(d) Proceeding as in part (b), we write

$$f(z) = \frac{1}{z-1}\,\frac{1}{1+(z-1)} = \frac{1}{(z-1)^2}\,\frac{1}{1+\dfrac{1}{z-1}}$$

$$= \frac{1}{(z-1)^2}\left[1 - \frac{1}{z-1} + \frac{1}{(z-1)^2} - \frac{1}{(z-1)^3} + \cdots\right]$$

$$= \frac{1}{(z-1)^2} - \frac{1}{(z-1)^3} + \frac{1}{(z-1)^4} - \frac{1}{(z-1)^5} + \cdots.$$

Because the series within the brackets converges for $|1/(z-1)| < 1$, the final series converges for $1 < |z-1|$.

## EXAMPLE 3 **Laurent Expansions**

Expand $f(z) = \dfrac{1}{(z-1)^2(z-3)}$ in a Laurent series valid for **(a)** $0 < |z-1| < 2$ and **(b)** $0 < |z-3| < 2$.

### Solution

(a) As in parts (c) and (d) of Example 2, we want only powers of $z-1$ and so we need to express $z-3$ in terms of $z-1$. This can be done by writing

$$f(z) = \frac{1}{(z-1)^2(z-3)} = \frac{1}{(z-1)^2}\,\frac{1}{-2+(z-1)} = \frac{-1}{2(z-1)^2}\,\frac{1}{1-\dfrac{z-1}{2}}$$

with the symbol $z$ replaced by $(z-1)/2$,

$$\begin{aligned} f(z) &= \frac{-1}{2(z-1)^2}\left[1 + \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} + \cdots\right] \\ &= -\frac{1}{2(z-1)^2} - \frac{1}{4(z-1)} - \frac{1}{8} - \frac{1}{16}(z-1) - \cdots . \end{aligned}$$

(b) To obtain powers of $z-3$, we write $z-1=2+(z-3)$ and

$$f(z)=\frac{1}{(z-1)^2(z-3)}=\frac{1}{z-3}\overbrace{[2+(z-3)]^{-2}}^{\substack{\text{We now factor 2}\\ \text{from this expression}}}$$

$$=\frac{1}{4(z-3)}\left[1+\frac{z-3}{2}\right]^{-2}$$

At this point we can obtain a power series for $\left[1+\dfrac{z-3}{2}\right]^{-2}$ by using the binomial expansion,†

$$f(z)=\frac{1}{4(z-3)}\left[1+\frac{(-2)}{1!}\left(\frac{z-3}{2}\right)+\frac{(-2)(-3)}{2!}\left(\frac{z-3}{2}\right)^2+\frac{(-2)(-3)(-4)}{3!}\left(\frac{z-3}{2}\right)^3+\cdots\right].$$

The binomial series in the brackets is valid for $|(z-3)/2|<1$ or $|z-3|<2$. Multiplying this series by $\dfrac{1}{4(z-3)}$ gives a Laurent series that is valid for $0<|z-3|<2$:

$$f(z)=\frac{1}{4(z-3)}-\frac{1}{4}+\frac{3}{16}(z-3)-\frac{1}{8}(z-3)^2+\cdots.$$

[†]For $\alpha$ real, the binomial series $(1+z)^{\alpha} = 1+\alpha z+\frac{\alpha(\alpha-1)}{2!}z^{2}+\frac{\alpha(\alpha-1)(\alpha-2)}{3!}z^{3}+\cdots$ is valid for $|z| < 1$.

### EXAMPLE 4 A Laurent Expansion

Expand $f(z) = \dfrac{8z+1}{z(1-z)}$ in a Laurent series valid for $0 < |z| < 1$.

**Solution** By partial fractions we can rewrite $f$ as

$$f(z) = \frac{8z+1}{z(1-z)} = \frac{1}{z} + \frac{9}{1-z}.$$

Then

$$\frac{9}{1-z} = 9 + 9z + 9z^2 + \cdots.$$

The foregoing geometric series converges for $|z| < 1$, but after we add the term $1/z$ to it, the resulting Laurent series

$$f(z) = \frac{1}{z} + 9 + 9z + 9z^2 + \cdots$$

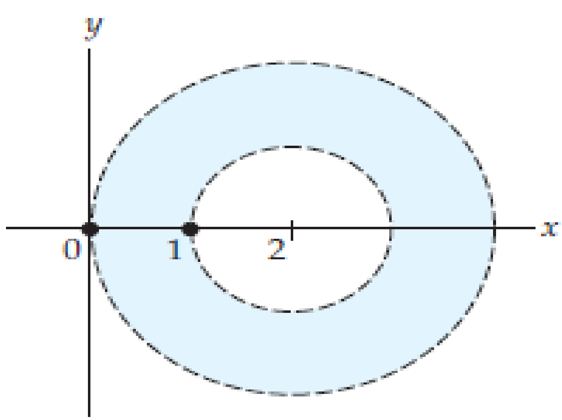
is valid for $0 < |z| < 1$.

## EXAMPLE 5 A Laurent Expansion

Expand $f(z) = \dfrac{1}{z(z-1)}$ in a Laurent series valid for $1 < |z-2| < 2$.

**Solution** The specified annular domain is shown in Figure  The center of this domain, $z = 2$, is the point of analyticity of the function $f$. Our goal now is to find two series involving integer powers of $z - 2$, one converging for $1 < |z-2|$ and the other converging for $|z-2| < 2$. To accomplish this, we proceed as in the last example by decomposing $f$ into partial fractions:

$$f(z) = -\frac{1}{z} + \frac{1}{z-1} = f_1(z) + f_2(z).$$ ***




Annular domain for Example 5

Now, $$f_1(z) = -\frac{1}{z} = -\frac{1}{2 + z - 2}$$

$$= -\frac{1}{2}\ \frac{1}{1 + \dfrac{z-2}{2}}$$

$$= -\frac{1}{2}\left[1 - \frac{z-2}{2} + \frac{(z-2)^2}{2^2} - \frac{(z-2)^3}{2^3} + \cdots\right]$$

$$= -\frac{1}{2} + \frac{z-2}{2^2} - \frac{(z-2)^2}{2^3} + \frac{(z-2)^3}{2^4} - \cdots$$

This series converges for $|(z-2)/2| < 1$ or $|z-2| < 2$. Furthermore,

$$f_2(z) = \frac{1}{z-1} = \frac{1}{1 + z - 2} = \frac{1}{z-2}\ \frac{1}{1 + \dfrac{1}{z-2}}$$

$$= \frac{1}{z-2}\left[1 - \frac{1}{z-2} + \frac{1}{(z-2)^2} - \frac{1}{(z-2)^3} + \cdots\right]$$

$$= \frac{1}{z-2} - \frac{1}{(z-2)^2} + \frac{1}{(z-2)^3} - \frac{1}{(z-2)^4} + \cdots$$

converges for $|1/(z-2)| < 1$ or $1 < |z-2|$. Substituting these two results in then gives

***

$$f(z) = \cdots - \frac{1}{(z-2)^4} + \frac{1}{(z-2)^3} - \frac{1}{(z-2)^2} + \frac{1}{z-2} - \frac{1}{2} + \frac{z-2}{2^2} - \frac{(z-2)^2}{2^3} + \frac{(z-2)^3}{2^4} - \cdots$$

This representation is valid for $z$ satisfying $|z-2| < 2$ and $1 < |z-2|$; in other words, for $1 < |z-2| < 2$.

## EXAMPLE 6 A Laurent Expansion

Expand $f(z) = e^{3/z}$ in a Laurent series valid for $0 < |z| < \infty$.

**Solution** we know that for all finite $z$, that is, $|z| < \infty$,

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots . \qquad *$$

We obtain the Laurent series for $f$ by simply replacing $z$ in * by $3/z$, $z \neq 0$,

$$e^{3/z} = 1 + \frac{3}{z} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots . \qquad **$$

This series ** is valid for $z \neq 0$, that is, for $0 < |z| < \infty$.

# Zeros and Poles

Suppose $z = z_0$ is an isolated singularity of a complex function $f$, and that

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k = \sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} + \sum_{k=0}^{\infty} a_k(z - z_0)^k \tag{1}$$

is the Laurent series representation of $f$ valid for the punctured open disk $0 < |z - z_0| < R$. We saw in the preceding section that a Laurent series (1) consists of two parts. That part of the series in (1) with negative powers of $z - z_0$, namely,

$$\sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} = \sum_{k=1}^{\infty} \frac{a_{-k}}{(z - z_0)^k} \tag{2}$$

is the **principal part** of the series. In the discussion that follows we will assign different names to the isolated singularity $z = z_0$ according to the number of terms in the principal part.

**Classification of Isolated Singular Points** An isolated singular point $z = z_0$ of a complex function $f$ is given a classification depending on whether the principal part (2) of its Laurent expansion (1) contains zero, a finite number, or an infinite number of terms.

(*i*) If the principal part is zero, that is, *all* the coefficients $a_{-k}$ in (2) are zero, then $z = z_0$ is called a **removable singularity**.

(*ii*) If the principal part contains a finite number of nonzero terms, then $z = z_0$ is called a **pole**. If, in this case, the last nonzero coefficient in (2) is $a_{-n}$, $n \geq 1$, then we say that $z = z_0$ is a **pole of order** $\boldsymbol{n}$. If $z = z_0$ is pole of order 1, then the principal part (2) contains exactly one term with coefficient $a_{-1}$. A pole of order 1 is commonly called a **simple pole**.

(*iii*) If the principal part (2) contains an infinitely many nonzero terms, then $z = z_0$ is called an **essential singularity**.

Table summarizes the form of a Laurent series for a function $f$ when $z = z_0$ is one of the above types of isolated singularities. Of course, $R$ in the table could be $\infty$.

| $z = z_0$ | Laurent Series for $0 < \lvert z - z_0 \rvert < R$ |
|---|---|
| Removable singularity | $a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$ |
| Pole of order $n$ | $\dfrac{a_{-n}}{(z - z_0)^n} + \dfrac{a_{-(n-1)}}{(z - z_0)^{n-1}} + \cdots + \dfrac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots$ |
| Simple pole | $\dfrac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$ |
| Essential singularity | $\cdots + \dfrac{a_{-2}}{(z - z_0)^2} + \dfrac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$ |

Table Forms of Laurent series

## EXAMPLE 1 Removable Singularity

by dividing the Maclaurin series for $\sin z$ by $z$, we see from

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots \tag{3}$$

that all the coefficients in the principal part of the Laurent series are zero. Hence $z = 0$ is a removable singularity of the function $f(z) = (\sin z)/z$.

## EXAMPLE 2 Poles and Essential Singularity

(a) Dividing the terms of $\sin z = z - \dfrac{z^3}{3!} + \dfrac{z^5}{5!} - \cdots$ by $z^2$ shows that

$$\frac{\sin z}{z^2} = \overbrace{\frac{1}{z}}^{\substack{\text{principal}\\ \text{part}}} - \frac{z}{3!} + \frac{z^3}{5!} - \cdots$$

for $0 < |z| < \infty$. From this series we see that $a_{-1} \neq 0$ and so $z = 0$ is a simple pole of the function $f(z) = (\sin z)/z^2$. In like manner, we see that $z = 0$ is a pole of order 3 of the function $f(z) = (\sin z)/z^4$

(b) we showed that the Laurent expansion of $f(z) = 1/(z-1)^2(z-3)$ valid for $0 < |z-1| < 2$ was

$$f(z) = \overbrace{-\frac{1}{2(z-1)^2} - \frac{1}{4(z-1)}}^{\text{principal part}} - \frac{1}{8} - \frac{z-1}{16} - \cdots.$$

Since $a_{-2} = -\frac{1}{2} \neq 0$, we conclude that $z = 1$ is a pole of order 2.

(c) the principal part of the Laurent expansion of the function $f(z) = e^{3/z}$ valid for $0 < |z| < \infty$ contains an infinite number of nonzero terms. This shows that $z = 0$ is an essential singularity of $f$.

**Zeros** Recall, a number $z_0$ is **zero** of a function $f$ if $f(z_0) = 0$. We say that an analytic function $f$ has a **zero of order** $n$ at $z = z_0$ if

$$\overbrace{f(z_0) = 0,\ f'(z_0) = 0,\ f''(z_0) = 0,\ \ldots,\ f^{(n-1)}(z_0) = 0}^{z_0 \text{ is a zero of } f \text{ and of its first } n-1 \text{ derivatives}},\ \text{but}\ f^{(n)}(z_0) \neq 0. \quad (4)$$

A zero of order $n$ is also referred to as a **zero of multiplicity** $n$. For example, for $f(z) = (z-5)^3$ we see that $f(5) = 0$, $f'(5) = 0$, $f''(5) = 0$, but $f'''(5) = 6 \neq 0$. Thus $f$ has a zero of order (or multiplicity) 3 at $z_0 = 5$. A zero of order 1 is called a **simple zero.**

The next theorem is a consequence of (4).

## Theorem Zero of Order $n$

A function $f$ that is analytic in some disk $|z - z_0| < R$ has a zero of order $n$ at $z = z_0$ if and only if $f$ can be written

$$f(z) = (z - z_0)^n \phi(z), \tag{5}$$

where $\phi$ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$.

## EXAMPLE 3 Order of a Zero

The analytic function $f(z) = z \sin z^2$ has a zero at $z = 0$. If we replace $z$ by $z^2$ in sin z we obtain the Maclaurin expansion

$$\sin z^2 = z^2 - \frac{z^6}{3!} + \frac{z^{10}}{5!} - \cdots.$$

Then by factoring $z^2$ out of the foregoing series we can rewrite $f$ as

$$f(z) = z \sin z^2 = z^3 \phi(z) \qquad \text{where} \qquad \phi(z) = 1 - \frac{z^4}{3!} + \frac{z^8}{5!} - \cdots \tag{6}$$

and $\phi(0) = 1$. When compared to (5), the result in (6) shows that $z = 0$ is a zero of order 3 of $f$.

## Theorem  Pole of Order $n$

A function $f$ analytic in a punctured disk $0 < |z - z_0| < R$ has a pole of order $n$ at $z = z_0$ if and only if $f$ can be written

$$f(z) = \frac{\phi(z)}{(z - z_0)^n}, \tag{7}$$

where $\phi$ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$.

**Theorem** **Pole of Order $n$**

If the functions $g$ and $h$ are analytic at $z = z_0$ and $h$ has a zero of order $n$ at $z = z_0$ and $g(z_0) \neq 0$, then the function $f(z) = g(z)/h(z)$ has a pole of order $n$ at $z = z_0$.

**Proof** Because the function $h$ has zero of order $n$, (5) gives $h(z) = (z - z_0)^n \phi(z)$, where $\phi$ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$. Thus $f$ can be written

$$f(z) = \frac{g(z)/\phi(z)}{(z - z_0)^n}. \qquad ****$$

Since $g$ and $\phi$ are analytic at $z = z_0$ and $\phi(z_0) \neq 0$, it follows that the function $g/\phi$ is analytic at $z_0$. Moreover, $g(z_0) \neq 0$ implies $g(z_0)/\phi(z_0) \neq 0$. We conclude from Theorem that the function $f$ has a pole of order $n$ at $z_0$.

When $n = 1$ in ****, we see that a zero of order 1, or a simple zero, in the denominator $h$ of $f(z) = g(z)/h(z)$ corresponds to a simple pole of $f$.

## EXAMPLE 4 Order of Poles

(a) Inspection of the rational function

$$f(z) = \frac{2z+5}{(z-1)(z+5)(z-2)^4}$$

shows that the denominator has zeros of order 1 at $z = 1$ and $z = -5$, and a zero of order 4 at $z = 2$. Since the numerator is not zero at any of these points, it follows from Theorem and **** that $f$ has simple poles at $z = 1$ and $z = -5$, and a pole of order 4 at $z = 2$.

(b) In Example 3 we saw that $z = 0$ is a zero of order 3 of $z \sin z^2$. From Theorem we conclude that the reciprocal function $f(z) = 1/(z \sin z^2)$ has a pole of order 3 at $z = 0$.

We saw in the last section that if a complex function $f$ has an *isolated singularity* at a point $z_0$, then $f$ has a Laurent series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k = \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots,$$

which converges for all $z$ *near* $z_0$. More precisely, the representation is valid in some deleted neighborhood of $z_0$ or punctured open disk $0 < |z - z_0| < R$. In this section our entire focus will be on the coefficient $a_{-1}$ and its importance in the evaluation of contour integrals.

**Residue** The coefficient $a_{-1}$ of $1/(z - z_0)$ in the Laurent series given above is called the **residue** of the function $f$ at the isolated singularity $z_0$. We shall use the notation

$$a_{-1} = \text{Res}(f(z),\ z_0)$$

to denote the residue of $f$ at $z_0$. Recall, if the principal part of the Laurent series valid for $0 < |z - z_0| < R$ contains a finite number of terms with $a_{-n}$ the last nonzero coefficient, then $z_0$ is a pole of order $n$; if the principal part of the series contains an infinite number of terms with nonzero coefficients, then $z_0$ is an essential singularity.

## EXAMPLE 1 Residues

(a) In part (b) of Example 2 we saw that $z = 1$ is a pole of order two of the function $f(z) = \dfrac{1}{(z-1)^2(z-3)}$. From the Laurent series obtained in that example valid for the deleted neighborhood of $z = 1$ defined by $0 < |z - 1| < 2$,

$$f(z) = \frac{-1/2}{(z-1)^2} + \overbrace{\frac{-1/4}{z-1}}^{a_{-1}} - \frac{1}{8} - \frac{z-1}{16} - \cdots$$

we see that the coefficient of $1/(z-1)$ is $a_{-1} = \operatorname{Res}(f(z), 1) = -\frac{1}{4}$.

(b) we saw that $z = 0$ is an essential singularity of $f(z) = e^{3/z}$. Inspection of the Laurent series obtained in that example,

$$e^{3/z} = 1 + \overbrace{\frac{3}{z}}^{a_{-1}} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots,$$

$0 < |z| < \infty$, shows that the coefficient of $1/z$ is $a_{-1} = \text{Res}(f(z), 0) = 3$.

**Theorem** **Residue at a Simple Pole**

If $f$ has a simple pole at $z = z_0$, then

$$\operatorname{Res}(f(z), z_0) = \lim_{z \to z_0} (z - z_0) f(z). \tag{1}$$

**Theorem** **Residue at a Pole of Order $n$**

If $f$ has a pole of order $n$ at $z = z_0$, then

$$\operatorname{Res}(f(z), z_0) = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} (z - z_0)^n f(z). \tag{2}$$

**Proof** Because $f$ is assumed to have pole of order $n$ at $z = z_0$, its Laurent expansion convergent on a punctured disk $0 < |z - z_0| < R$ must have the form

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \cdots + \frac{a_{-2}}{(z-z_0)^2} + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \cdots\,,$$

where $a_{-n} \neq 0$. We multiply the last expression by $(z - z_0)^n$,

$$(z-z_0)^n f(z) = a_{-n} + \cdots + a_{-2}(z-z_0)^{n-2} + a_{-1}(z-z_0)^{n-1} + a_0(z-z_0)^n + a_1(z-z_0)^{n+1} + \cdots$$

and then differentiate both sides of the equality $n - 1$ times:

$$\frac{d^{n-1}}{dz^{n-1}}(z-z_0)^n f(z) = (n-1)!a_{-1} + n!a_0(z-z_0) + \cdots\,. \tag{3}$$

Since all the terms on the right-hand side after the first involve positive integer powers of $z - z_0$, the limit of (3) as $z \to z_0$ is

$$\lim_{z\to z_0} \frac{d^{n-1}}{dz^{n-1}}(z-z_0)^n f(z) = (n-1)!a_{-1}.$$

Solving the last equation for $a_{-1}$ gives (2).

Notice that (2) reduces to (1) when $n = 1$.

**EXAMPLE 2 Residue at a Pole**

The function $f(z) = \dfrac{1}{(z-1)^2(z-3)}$ has a simple pole at $z = 3$ and a pole of order 2 at $z = 1$. Use Theorems to find the residues.

**Solution** Since $z = 3$ is a simple pole, we use (1):

$$\operatorname{Res}(f(z), 3) = \lim_{z\to 3}(z-3)f(z) = \lim_{z\to 3}\frac{1}{(z-1)^2} = \frac{1}{4}.$$

Now at the pole of order 2, the result in (2) gives

$$\operatorname{Res}(f(z), 1) = \frac{1}{1!}\lim_{z\to 1}\frac{d}{dz}(z-1)^2 f(z)$$

$$= \lim_{z\to 1}\frac{d}{dz}\frac{1}{z-3}$$

$$= \lim_{z\to 1}\frac{-1}{(z-3)^2} = -\frac{1}{4}.$$

When $f$ is not a rational function, calculating residues by means of (1) or (2) can sometimes be tedious. It is possible to devise alternative residue formulas. In particular, suppose a function $f$ can be written as a quotient $f(z) = g(z)/h(z)$, where $g$ and $h$ are analytic at $z = z_0$. If $g(z_0) \neq 0$ and if the function $h$ has a zero of order 1 at $z_0$, then $f$ has a simple pole at $z = z_0$ and

$$\mathrm{Res}(f(z), z_0) = \frac{g(z_0)}{h'(z_0)}. \tag{4}$$

To derive this result we shall use the definition of a zero of order 1, the definition of a derivative, and then (1). First, since the function $h$ has a zero of order 1 at $z_0$, we must have $h(z_0) = 0$ and $h'(z_0) \neq 0$. Second, by definition of the derivative

$$h'(z_0) = \lim_{z \to z_0} \frac{h(z) - \overbrace{h(z_0)}^{0}}{z - z_0} = \lim_{z \to z_0} \frac{h(z)}{z - z_0}.$$

We then combine the preceding two facts in the following manner in (1):

$$\operatorname{Res}(f(z), z_0) = \lim_{z \to z_0} (z - z_0) \frac{g(z)}{h(z)} = \lim_{z \to z_0} \frac{g(z)}{\dfrac{h(z)}{z - z_0}} = \frac{g(z_0)}{h'(z_0)}.$$

## EXAMPLE 3 Using (4) to Compute Residues

The polynomial $z^4+1$ can be factored as $(z-z_1)(z-z_2)(z-z_3)(z-z_4)$, where $z_1$, $z_2$, $z_3$, and $z_4$ are the four distinct roots of the equation $z^4+1=0$

(or equivalently, the four fourth roots of $-1$). It follows from Theorem that the function

$$f(z)=\frac{1}{z^4+1}$$

has four simple poles. we have $z_1=e^{\pi i/4}$, $z_2=e^{3\pi i/4}$, $z_3=e^{5\pi i/4}$, and $z_4=e^{7\pi i/4}$. To compute the residues, we use (4) of this section along with Euler's formula

$$\operatorname{Res}(f(z),z_1)=\frac{1}{4z_1^3}=\frac{1}{4}e^{-3\pi i/4}=-\frac{1}{4\sqrt{2}}-\frac{1}{4\sqrt{2}}i$$

$$\operatorname{Res}(f(z),z_2)=\frac{1}{4z_2^3}=\frac{1}{4}e^{-9\pi i/4}=\frac{1}{4\sqrt{2}}-\frac{1}{4\sqrt{2}}i$$

$$\operatorname{Res}(f(z),z_3)=\frac{1}{4z_3^3}=\frac{1}{4}e^{-15\pi i/4}=\frac{1}{4\sqrt{2}}+\frac{1}{4\sqrt{2}}i$$

$$\operatorname{Res}(f(z),z_4)=\frac{1}{4z_4^3}=\frac{1}{4}e^{-21\pi i/4}=-\frac{1}{4\sqrt{2}}+\frac{1}{4\sqrt{2}}i.$$

Of course, we could have calculated each of the residues in Example 3 using formula (1).

$$\operatorname{Res}(f(z), z_1) = \lim_{z \to z_1} (z - z_1) \frac{1}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)}$$

$$= \frac{1}{(z_1 - z_2)(z_1 - z_3)(z_1 - z_4)}$$

$$= \frac{1}{(e^{\pi i/4} - e^{3\pi i/4})(e^{\pi i/4} - e^{5\pi i/4})(e^{\pi i/4} - e^{7\pi i/4})}.$$

Finally, we must do this process three more times.

**Theorem** Cauchy's Residue Theorem

Let $D$ be a simply connected domain and $C$ a simple closed contour lying entirely within $D$. If a function $f$ is analytic on and within $C$, except at a finite number of isolated singular points $z_1, z_2, \ldots, z_n$ within $C$, then

$$\oint_C f(z)\,dz = 2\pi i \sum_{k=1}^{n} \operatorname{Res}\left(f(z), z_k\right). \tag{5}$$

## EXAMPLE 4 Evaluation by the Residue Theorem

Evaluate $\oint_C \frac{1}{(z-1)^2(z-3)}\,dz$, where

(a) the contour $C$ is the rectangle defined by $x=0$, $x=4$, $y=-1$, $y=1$,

(b) and the contour $C$ is the circle $|z|=2$.

### Solution

(a) Since both $z=1$ and $z=3$ are poles within the rectangle we have from (5) that

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz = 2\pi i\,[\operatorname{Res}(f(z),1)+\operatorname{Res}(f(z),3)]$$

We found these residues in Example 2. Therefore,

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz = 2\pi i\left[\left(-\frac{1}{4}\right)+\frac{1}{4}\right]=0.$$

(b) Since only the pole $z=1$ lies within the circle $|z|=2$, we have from (5)

$$\oint_C \frac{1}{(z-1)^2(z-3)}\,dz = 2\pi i\,\operatorname{Res}(f(z),1) = 2\pi i\left(-\frac{1}{4}\right) = -\frac{\pi}{2}i.$$

## EXAMPLE 5 Evaluation by the Residue Theorem

Evaluate $\oint_C \frac{2z+6}{z^2+4}\,dz$, where the contour $C$ is the circle $|z-i|=2$.

**Solution** By factoring the denominator as $z^2+4=(z-2i)(z+2i)$ we see that the integrand has simple poles at $-2i$ and $2i$. Because only $2i$ lies within the contour $C$, it follows from (5) that

$$\oint_C \frac{2z+6}{z^2+4}\,dz = 2\pi i\,\mathrm{Res}(f(z),2i).$$

But

$$\mathrm{Res}(f(z),2i) = \lim_{z\to 2i}(z-2i)\frac{2z+6}{(z-2i)(z+2i)}$$

$$= \frac{6+4i}{4i} = \frac{3+2i}{2i}.$$

Hence,

$$\oint_C \frac{2z+6}{z^2+4}\,dz = 2\pi i\left(\frac{3+2i}{2i}\right) = \pi(3+2i).$$

## EXAMPLE 6 Evaluation by the Residue Theorem

Evaluate $\oint_C \frac{e^z}{z^4+5z^3}\,dz$, where the contour $C$ is the circle $|z|=2$.

**Solution** Writing the denominator as $z^4+5z^3=z^3(z+5)$ reveals that the integrand $f(z)$ has a pole of order 3 at $z=0$ and a simple pole at $z=-5$. But only the pole $z=0$ lies within the given contour and so from (5) and (2) we have,

$$\oint_C \frac{e^z}{z^4+5z^3}\,dz = 2\pi i\,\mathrm{Res}(f(z),0) = 2\pi i\,\frac{1}{2!}\lim_{z\to 0}\frac{d^2}{dz^2}z^3\cdot\frac{e^z}{z^3(z+5)}$$

$$= \pi i \lim_{z\to 0}\frac{(z^2+8z+17)e^z}{(z+5)^3} = \frac{17\pi}{125}i.$$

## EXAMPLE 7 Evaluation by the Residue Theorem

Evaluate $\oint_C \tan z\, dz$, where the contour $C$ is the circle $|z| = 2$.

**Solution** The integrand $f(z) = \tan z = \sin z/\cos z$ has simple poles at the points where $\cos z = 0$. the only zeros $\cos z$

are the real numbers $z = (2n+1)\pi/2$, $n = 0, \pm 1, \pm 2, \ldots$ . Since only $-\pi/2$ and $\pi/2$ are within the circle $|z| = 2$, we have

$$\oint_C \tan z\, dz = 2\pi i\left[\operatorname{Res}\left(f(z), -\frac{\pi}{2}\right) + \operatorname{Res}\left(f(z), \frac{\pi}{2}\right)\right].$$

With the identifications $g(z) = \sin z$, $h(z) = \cos z$, and $h'(z) = -\sin z$, we see from (4) that

$$\operatorname{Res}\left(f(z), -\frac{\pi}{2}\right) = \frac{\sin(-\pi/2)}{-\sin(-\pi/2)} = -1$$

and
$$\operatorname{Res}\left(f(z), \frac{\pi}{2}\right) = \frac{\sin(\pi/2)}{-\sin(\pi/2)} = -1.$$

Therefore,
$$\oint_C \tan z\, dz = 2\pi i[-1-1] = -4\pi i.$$

## EXAMPLE 8 Evaluation by the Residue Theorem

Evaluate $\oint_C e^{3/z}\,dz$, where the contour $C$ is the circle $|z| = 1$.

**Solution** As we have seen, $z = 0$ is an essential singularity of the integrand $f(z) = e^{3/z}$ and so neither formulas (1) and (2) are applicable to find the residue of $f$ at that point. Nevertheless, we saw in Example 1 that the Laurent series of $f$ at $z = 0$ gives $\operatorname{Res}(f(z),\ 0) = 3$. Hence from (5) we have

$$\oint_C e^{3/z}\,dz = 2\pi i\,\operatorname{Res}(f(z), 0) = 2\pi i\,(3) = 6\pi i.$$